Distance between two compact sets
Pugh, 2.46

Theorem 1. *Let M be a metric space. Assume that A and B are compact, disjoint, nonempty subsets of M. Prove that there are $a_0 \in A$ and $b_0 \in B$ such that $d(a_0, b_0) \leq d(a, b)$ for all $a \in A$ and $b \in B$.*

Proof. Consider the product set

$$A \times B = \{(a, b) : a \in A,\ b \in B\} \subseteq M \times M.$$

Since A and B are compact, their product $A \times B$ is compact. Since A and B are nonempty, the product $A \times B$ is also nonempty.

Let $f : M \times M \to \mathbb{R}$ be the distance function defined by

$$f(x, y) = d(x, y).$$

This function is continuous on $M \times M$, and hence it is continuous on $A \times B$.

A continuous real-valued function on a nonempty compact set attains its minimum. Therefore, there exists a point $(a_0, b_0) \in A \times B$ at which f attains its minimum on $A \times B$. In particular, $a_0 \in A$ and $b_0 \in B$, and

$$f(a_0, b_0) \leq f(a, b) \qquad \text{for all } (a, b) \in A \times B.$$

Now let $a \in A$ and $b \in B$ be arbitrary. Then $(a, b) \in A \times B$, so the minimality of (a_0, b_0) gives

$$d(a_0, b_0) = f(a_0, b_0) \leq f(a, b) = d(a, b).$$

Thus $a_0 \in A$ and $b_0 \in B$ satisfy $d(a_0, b_0) \leq d(a, b)$ for all $a \in A$ and $b \in B$, as required. □